EXPERIMENT NUMBER: 3

MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

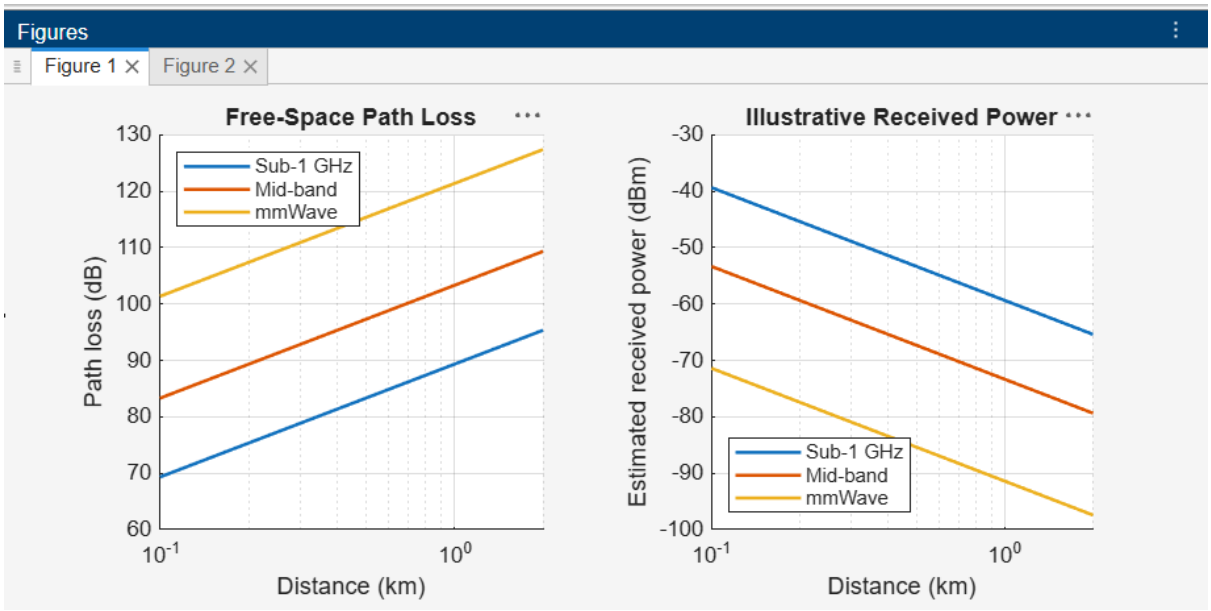
Matlab Program

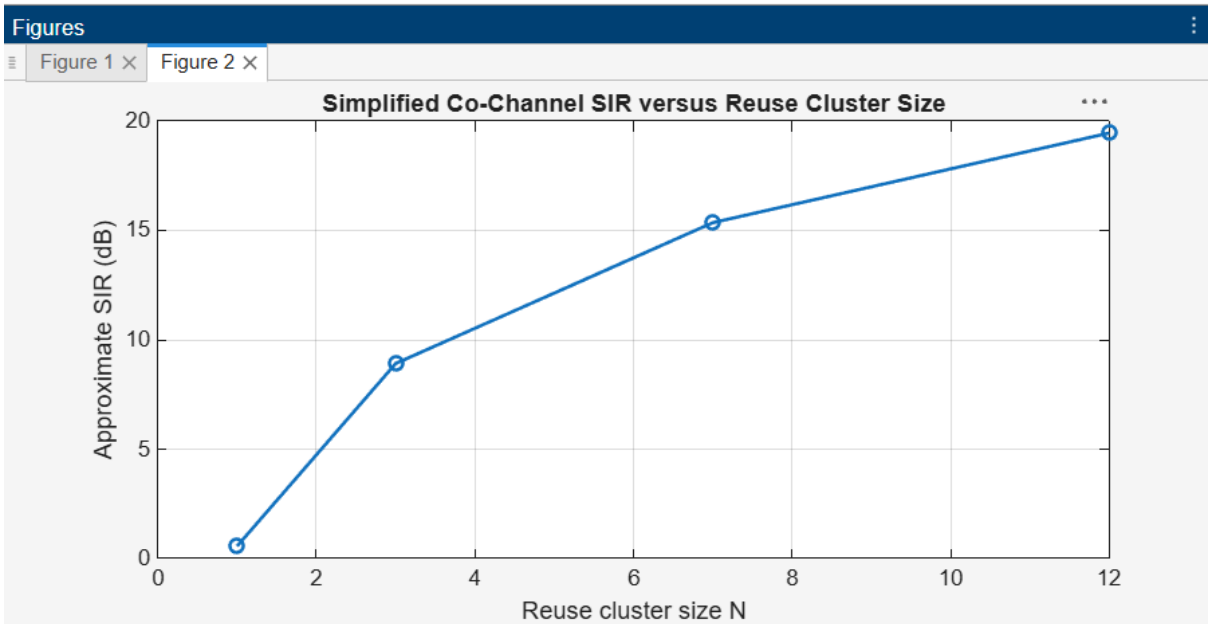
```
clear;
clc;
close all;
% Representative bands and link-budget assumptions
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0;
grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
% Free-space path loss (d in km, f in MHz)
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
% Path loss and received power over distance
figure;
tiledlayout(1,2);
nexttile;
hold on;
grid on;
for k=1:numel(freqMHz)
```

```
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log');
xlabel('Distance (km)');
ylabel('Path loss (dB)');
legend(bandName,'Location','northwest');
title('Free-Space Path Loss');
nexttile;
hold on;
grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log');
xlabel('Distance (km)');
ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest');
title('Illustrative Received Power');
% Simple first-tier co-channel reuse estimate
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
    'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure;
plot(reuseN,sirDb,'o-','LineWidth',1.5);
```

```
grid on;
xlabel('Reuse cluster size N');
ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

Result





Command Window

Band	Frequency_GHz	FSPL_dB	EstimatedRx_dBm
"Sub-1 GHz"	0.7	83.321	-53.321
"Mid-band"	3.5	97.301	-67.301
"mmWave"	28	115.36	-85.363

ReuseClusterN	ApproxInterfererDistance_km	ApproxSIR_dB
1	0.86603	0.56811
3	1.5	8.9177
7	2.2913	15.357
12	3	19.454

At 0.50 km, estimated received powers are:

Band	Rx_dBm
"Sub-1 GHz"	-53.321
"Mid-band"	-67.301
"mmWave"	-85.363

>>